

Jan Hlavsa

Curriculum Vitae

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Born: 3 November 2002



Education

- 2025 – **Ing. in Open Informatics, specialization in Artificial Intelligence**, *Faculty of Electrical Engineering, CTU Prague*, Prague, Czech Republic
expected 2027 The master's program focused on artificial intelligence, algorithms, and applied computer science.
- 2022 – 2025 **Bc. in Open Informatics, specialization in Artificial Intelligence**, *Faculty of Electrical Engineering, CTU Prague*, Prague, Czech Republic
Bachelor's program focused on artificial intelligence, algorithms, and applied computer science. Graduated with a graded weighted average of 1.0.
- 2014 – 2022 **High School Diploma**, *Gymnázium Jana Palacha*, Mělník, Czech Republic

Work Experience

- July – August 2026 **Student Summer Research Fellow (SSRF)**, *ETH Zurich, Computer Vision and Geometry Group*, Zurich, Switzerland, *Group head: Prof. Marc Pollefeys; Supervisors: Dr. Daniel Barath and Dr. Zuria Bauer*
Selected as one of 20 fellows from over 2,500 applicants ($< 0.8\%$ acceptance rate). Conducted research on *Improving Global-Aware Edge Prioritization for Pose Graph Initialization through (k, ℓ) -Sparse Graphs*. Continuing active collaboration on the project remotely alongside ongoing research at CTU.
- 2026 – **Teaching Assistant**, *FEE CTU, Department of Cybernetics*, Prague, Czech Republic
Present Teaching tutorials for the Optimization course (Winter Semester 2026/27, Summer Semester 2025/26), supervised by Doc. Tomáš Werner.
- 2025 – **Teaching Assistant**, *FEE CTU, Department of Cybernetics*, Prague, Czech Republic
Present Teaching tutorials on deep learning and convolutional neural networks for the Deep Learning Essentials course (formerly Robotics; Winter Semesters 2025/26, 2026/27), supervised by Doc. Karel Zimmermann.
- 2023 – **Intern**, *FEE CTU, Visual Recognition Group*, Prague, Czech Republic, *Supervisor: Prof. Ing. Jiří Matas, PhD. and Ing. Bc. Radim Špetlík, PhD.*
Present Engaged in computer vision and data science research. Currently leading multidisciplinary research combining GC×GC-TOFMS data analysis with deep learning, and continuing research on pose graph optimization in close collaboration with Prof. Jiří Matas and Dr. Daniel Barath & Dr. Zuria Bauer (ETH Zurich).

Academic Service

- 2026 – **Academic Senator**, *Academic Senate of FEE CTU*, Prague, Czech Republic
Present Elected student representative in the Academic Senate of the Faculty of Electrical Engineering, CTU Prague.

Publications

- MVA (under review) **Identity Verification from Human Scent using Learned Representations of Two-Dimensional Gas Chromatography–Mass Spectrometry Data**
Hlavsa J., Špetlík R., Čechová J., Pojmanová P., Matas J., Urban Š.
Manuscript submitted to Machine Vision and Applications (MVA); currently under review.

- WACV 2026 **Identity Verification from Human Scent using Channel Representation of 2D Gas Chromatography-Mass Spectrometry Data**
 Spetlik R., Hlavsa J., Čechová J., Pojmanová P., Matas J., Urban Š.
Peer-reviewed paper accepted to the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2026.
- SCIA 2025 **Sex Classification from Human Scent Using Image Interpretation of 2D Gas Chromatography-Mass Spectrometry Data**
 Hlavsa J., Spetlik R., Čechová J., Pojmanová P., Matas J., Urban Š.
Peer-reviewed paper accepted for presentation at the Scandinavian Conference on Image Analysis (SCIA) 2025 and published in the conference proceedings.

Conferences & Courses

- 2026 **Oral & Poster Presentation**, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Tucson – Presented research on identity verification using GC×GC-TOFMS data.
- 2024 – 2025 **Prg.ai Minor**, completion of the prg.ai minor program, which combines the best of computer science courses from CTU and Charles University.
- 2025 **Poster**, Scandinavian Conference on Image Analysis (SCIA), Reykjavík - Poster presentation of our research focused on sex classification and identity verification from human scent.
- 2025 **Speaker**, The Multidimensional Chromatography Workshop, Liège - Short oral presentation about the application of computer vision methods to GC×GC-TOFMS data.
- 2024 **Participant**, Vision and Sports Summer School (organized by Vittorio Ferrari – Synthesia, Ondřej Chum and Jiří Matas – CTU Prague).

Awards

- **Dean's Award (2025)** — Awarded by the Dean of FEE CTU for an outstanding bachelor's thesis on the use of GC×GC-TOFMS data of human scent to predict sex and verify identity.
- **Winner of Rakathon (2025)** — Member of the winning team at the hackathon against cancer organized by the Second Faculty of Medicine, Charles University, and Motol University Hospital. Developed an AI pipeline for registration, alignment, and tumor progression tracking from CT scans.

Skills

Programming

- **Python**: Advanced
- **C**: Advanced
- **Julia**: Intermediate
- **C++**: Intermediate
- **Java**: Intermediate
- **C#**: Intermediate
- **PostgreSQL**: Intermediate
- **Haskell**: Beginner

Data Science Machine Learning, Deep Learning (PyTorch), Computer Vision, Data Analysis, High-Performance Computing

Tools Linux, Git, VS Code, LaTeX, JetBrains Toolbox, Microsoft Office

Languages

Czech Native
 English Fluent

French Intermediate

Additional Information

Interests Saxophone, Taekwondo

Memberships Member of Upsilon Pi Epsilon (UPE) Honor Society